



Written exam
„Numerical Optimization and Model Predictive Control“
on

Please fill out legibly:

Last name, first name:

Student registration number:

I hereby declare that I am eligible for examination and have read the instructions on page 3.

Signature:

Total number of printed sheets: 5

Good luck!

Question	1	2	3	4	Sum	Grade
Points	/ 20	/ 25	/ 30	/ 15	/ 90	

Information about the exam

Evaluation of the exam

- If you are not on the list of registered students, your participation at the exam is considered with reservation.
- Results without comprehensible argumentation or without clear mathematical derivation cannot be accounted for.
- Use only document-proof pens. Do not use red color.

Notes on the exam

- Subtasks marked with * are new tasks (no follow-up), whereas subtasks without * build upon previous ones.
- Answer the subtasks in the prepared areas. Use the enclosed spare pages if there is not enough space. In any case, mark all subtasks with the task number (e.g. „1c“).
- Please do not remove the staples.

Tools

- The lecture notes, the exercise materials and transcripts in printed form can be used in the exam. Also computer-written notes, summaries and formularies with clear reference to the course may be used in printed form.
- Previous exams, books, calculators and materials from other courses are not permitted.

Question 1 (20 Points)

A component manufactured in two supplier plants S_1 and S_2 is to be distributed to two processing plants P_1 and P_2 via the transport network shown in Figure 1 consisting of four transport routes $t_1, \dots, t_4$. The two processing plants have constant demands D_1 and D_2 for the delivered component. The transport costs over the four different routes are directly proportional to the transported quantity x_i with the proportionality factor C_i .

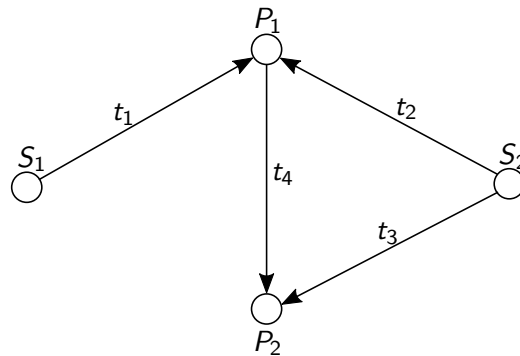


Figure 1: Structure of the transport network.

The task is to find the optimal distribution of the transport quantities x_i to minimize the transport costs while guaranteeing the demand D_1 and D_2 of the processing plants. The transport quantities over the individual routes and the production volume of the suppliers are unbounded from above. Consider the values $D_1 = 20$ and $D_2 = 10$ for the demand and $C_1 = 1$, $C_2 = 2$, $C_3 = 10$ and $C_4 = 0.1$ as factors for the transport costs.

- (4 P) a) Formulate the transportation problem as a linear optimization problem in standard form.
- (4 P) b) Formulate the KKT conditions for the optimization problem.
- (4 P) c) In the following, the transportation problem is to be solved using the simplex algorithm shown in Table 1. Examine for each of the following combinations

$$\mathcal{B}_1 = \{1, 2\}$$

$$\mathcal{B}_2 = \{1, 3\}$$

$$\mathcal{B}_3 = \{3, 4\}$$

whether it is an basic feasible solution.

- (8 P) d) Solve the transportation problem using the simplex method. Start with an basic feasible solution from problem c).

Question 2 (25 Points)

Consider the constrained nonlinear optimization problem

$$\min_{\mathbf{x} \in \mathbb{R}^2} k_1 x_1^3 + k_2 x_1 x_2 + k_3 x_2^5 \quad (1a)$$

$$\text{s.t.} \quad -\frac{1}{2}x_1 x_2 - \frac{1}{2}x_2 + \frac{7}{8} \leq 0 \quad (1b)$$

$$-\frac{1}{2}x_1 + x_2 - \frac{5}{8} \leq 0 \quad (1c)$$

$$\frac{1}{2}x_1^3 x_2 - \frac{7}{2} \leq 0, \quad (1d)$$

which is to be solved in the following with the help of the SQP method shown in table 2. Therefore, first the linear-quadratic approximation

$$\min_{\mathbf{s} \in \mathbb{R}^n} \frac{1}{2} \mathbf{s}^T \nabla_{\mathbf{x}\mathbf{x}}^2 L(\mathbf{x}^k, \boldsymbol{\lambda}^k, \boldsymbol{\mu}^k) \mathbf{s} + \nabla f(\mathbf{x}^k)^T \mathbf{s} \quad (2a)$$

$$\text{s.t.} \quad \nabla g_i(\mathbf{x}^k)^T \mathbf{s} + g_i(\mathbf{x}^k) = 0, \quad i = 1, \dots, p \quad (2b)$$

$$\nabla h_i(\mathbf{x}^k)^T \mathbf{s} + h_i(\mathbf{x}^k) \leq 0, \quad i = 1, \dots, q, \quad (2c)$$

of the nonlinear problem should be formed at the current iteration point $\mathbf{x}^k$.

- State the Lagrangian function for the nonlinear problem and use it to compute the linear-quadratic approximation at the iteration point $\mathbf{x}^k = [1, 1]^T$, $\boldsymbol{\mu}^k = [0, 0, 0]^T$ depending on the parameters k_1 , k_2 , and k_3 . (9 P)
- The contour lines of the quadratic cost function (2a) are shown in Figure 2. Draw the admissible set of the problem (2) into the figure and use it to determine the solution of the approximation graphically. Then compute the new iteration point $\mathbf{x}^{k+1}$. (9 P)
- An alternative to the graphical solution of the underlying quadratic problem is the interior point method. Derive the interior point method for a quadratic problem of the form (7 P)

$$\begin{aligned} \min_{\mathbf{x} \in \mathbb{R}^n} \quad & \frac{1}{2} \mathbf{x}^T \mathbf{G} \mathbf{x} + \mathbf{c}^T \mathbf{x} \\ \text{s.t.} \quad & \mathbf{A} \mathbf{x} = \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0}. \end{aligned}$$

What is the role of the parameter τ^k and how is it adapted? Which system of equations must be solved to calculate the search direction $\mathbf{s}^k$? Which conditions have to be considered when determining the step size α^k ?

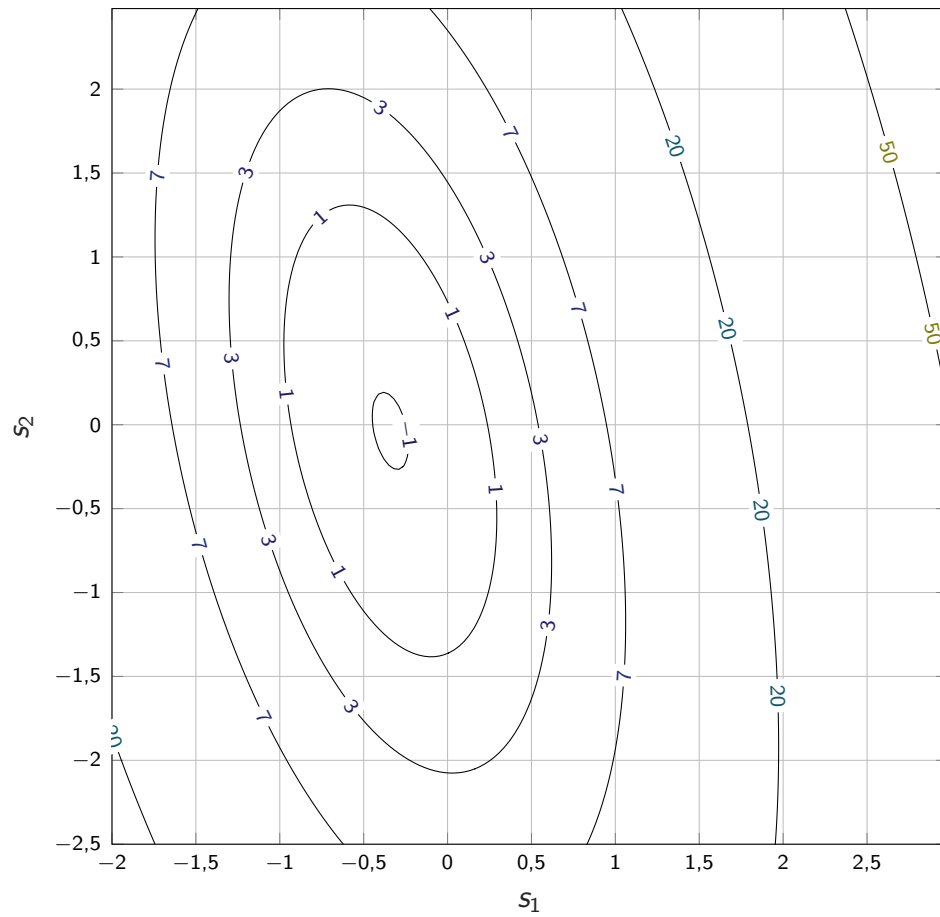


Figure 2: Contour lines of the linear-quadratic approximation of Question 2.

Question 3 (30 Points)

Consider the following optimal control problem

$$\begin{aligned} \min_u \quad & \int_0^T \frac{1}{2} q x(t)^2 + \frac{1}{2} r u(t)^2 dt \\ \text{s.t.} \quad & \dot{x}(t) = ax(t) + bu(t), \quad x(0) = x_0 \end{aligned}$$

with the state $x(t) \in \mathbb{R}$, the control input $u(t) \in \mathbb{R}$, the weights $q > 0$, $r > 0$ in the cost functional, the parameters a , b in the system dynamics, the time horizon $T > 0$ and the initial value x_0 .

- Establish the corresponding Hamiltonian function $H(x, u, \lambda)$ and derive the necessary first-order optimality conditions. (3 P)
- From the optimality conditions, determine the optimal solution $x^*(t)$, $u^*(t)$, and $\lambda^*(t)$. (10 P)
Note: A differential equation of the form $\ddot{x}(t) = sx(t)$ has the solution $x(t) = c_1 e^{\sqrt{s}t} + c_2 e^{-\sqrt{s}t}$ with $c_1, c_2 \in \mathbb{R}$.
- In terms of model predictive control, the problem is solved optimally in each sampling step and the start of the optimal control input $u^*(t)$ for $t \in [0, \Delta t]$ is applied to asymptotically stabilize the equilibrium $(x_e, u_e) = (0, 0)$. What is the requirement for the horizon length T in dependence of the parameters a , b , q , r and the sampling time $\Delta t > 0$, so that the condition (10 P)

$$|x^*(\Delta t)| < |x_0|$$

is fulfilled in every sampling step?

- Alternatively, the optimal control problem can be extended by a terminal cost $\frac{1}{2} p x(T)^2$. Compute a suitable weight $p > 0$ using the algebraic Riccati equation, which in the general case (4 P)

$$\begin{aligned} \min_u \quad & \int_0^T \frac{1}{2} \mathbf{x}(t)^\top \mathbf{Q} \mathbf{x}(t) + \frac{1}{2} \mathbf{u}(t)^\top \mathbf{R} \mathbf{u}(t) dt \\ \text{s.t.} \quad & \dot{\mathbf{x}}(t) = \mathbf{A} \mathbf{x}(t) + \mathbf{B} \mathbf{u}(t), \quad \mathbf{x}(0) = \mathbf{x}_0 \end{aligned}$$

is given by

$$\mathbf{P} \mathbf{A} + \mathbf{A}^\top \mathbf{P} - \mathbf{P} \mathbf{B} \mathbf{R}^{-1} \mathbf{B}^\top \mathbf{P} + \mathbf{Q} = \mathbf{0}.$$

What are the advantages and disadvantages of this approach compared to a sufficiently long horizon?

- Calculate the optimal control input $u^*(t)$ in dependence of $x^*(t)$ and $\lambda^*(t)$ for the case of additional input constraints $u(t) \in [u^-, u^+]$ in the above optimal control problem. To this end, apply Pontryagin's maximum principle (3 P)

$$H(x^*(t), u^*, \lambda^*(t)) = \min_{u \in [u^-, u^+]} H(x^*(t), u, \lambda^*(t)) \quad \forall t \in [0, T].$$

Note: It is not necessary to explicitly state $x^*(t)$ and $\lambda^*(t)$ for this case.

Question 4 (15 Points)

(2 P) * a) Under what conditions is a quasi-Newton method guaranteed to converge in at most n iterations for an optimization problem $\min_{\mathbf{x}} f(\mathbf{x})$ with $\mathbf{x} \in \mathbb{R}^n$?

(4 P) * b) For solving nonlinear optimization problems with sequential quadratic programming (SQP), typically a penalty function is used to determine the step size. Why is this necessary in SQP methods and not in the Active set method for solving quadratic optimization problems? Why is the l_1 -penalty function

$$P(\mathbf{x}, \sigma) = f(\mathbf{x}) + \sigma \left[\sum_{i=1}^p |g_i(\mathbf{x})| + \sum_{i=1}^q \max\{0, h_i(\mathbf{x})\} \right]$$

called an exact penalty function in this context?

(4 P) * c) Consider the optimal control problem

$$\begin{aligned} \min_u \quad & \int_0^T l(x(t)) + ru(t) dt \\ \text{s.t.} \quad & \dot{x}(t) = f(x(t)) + g(x(t))u(t), \quad x(0) = x_0 \end{aligned}$$

with the state $x(t) \in \mathbb{R}$, the control input $u(t) \in \mathbb{R}$, the time horizon T and the initial value x_0 . The partial derivative of the Hamiltonian function with respect to u is in this case

$$\frac{\partial H(x, u, \lambda)}{\partial u} = r + \lambda g(x)$$

with the adjoint state λ and thus does not depend directly on u . What is this case called and what possibilities exist to compute the optimal control for it?

(5 P) * d) Given is the following optimal control problem

$$\begin{aligned} \min_u \quad & \frac{1}{2} \mathbf{x}(T)^T \mathbf{S} \mathbf{x}(T) + \frac{1}{2} \int_0^T \mathbf{x}(t)^T \mathbf{Q} \mathbf{x}(t) + \mathbf{u}(t)^T \mathbf{R} \mathbf{u}(t) dt \\ \text{s.t.} \quad & \dot{\mathbf{x}}(t) = \mathbf{A} \mathbf{x}(t) + \mathbf{B} \mathbf{u}(t), \quad \mathbf{x}(t) = \mathbf{x}_0 \\ & \mathbf{C} \mathbf{x}(t) \leq \mathbf{d}, \quad t \in [0, T] \\ & \mathbf{u}(t) \in [\mathbf{u}^-, \mathbf{u}^+], \quad t \in [0, T] \end{aligned}$$

with the state $\mathbf{x}(t) \in \mathbb{R}^n$, the control input $\mathbf{u}(t) \in \mathbb{R}^m$, the weighting matrices $\mathbf{S}$, $\mathbf{Q}$, $\mathbf{R}$, the time horizon T and the initial value $\mathbf{x}_0$. This problem is to be solved numerically using a full discretization. Give the resulting static optimization problem when using the trapezoidal rule

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{f}(\mathbf{x}(t)) \quad \rightarrow \quad \frac{\mathbf{x}^{k+1} - \mathbf{x}^k}{t^{k+1} - t^k} = \frac{1}{2} (\mathbf{f}(\mathbf{x}^k) + \mathbf{f}(\mathbf{x}^{k+1}))$$

with $t^k = k\Delta t$ and $\mathbf{x}^k \approx \mathbf{x}(t^k)$. How many optimization variables result in dependence of the $N + 1$ grid points? Which methods can be considered for the numerical solution of the static problem?

Given (by initialization or previous iteration step):

$$\mathcal{B}, \mathcal{N}, \mathbf{x}_B = \mathbf{B}^{-1}\mathbf{b} \geq \mathbf{0}, \mathbf{x}_N = \mathbf{0}$$

KKT conditions:

$$\boldsymbol{\lambda} \leftarrow -(\mathbf{B}^\top)^{-1}\mathbf{c}_B$$

Lagrange multipliers

$$\boldsymbol{\mu}_N \leftarrow \mathbf{c}_N + \mathbf{N}^\top \boldsymbol{\lambda}$$

if $\boldsymbol{\mu}_N \geq \mathbf{0}$ **then**

$$\text{return } \mathbf{x}_B^* = \mathbf{x}_B, \mathbf{x}_N^* = \mathbf{x}_N$$

Optimal solution

end

Next basic feasible solution:

$$\text{Index } s \in \mathcal{N} \text{ with } \mu_s < 0 \text{ from } \mu_s = \min_{i \in \mathcal{N} | \mu_i < 0} \mu_i$$

$$\text{Direction } \mathbf{d} = \mathbf{B}^{-1}\mathbf{A}_s$$

if $\mathbf{d} \leq \mathbf{0}$ **then**

stop

Unconstrained problem

end

$$x_s^+ \text{ and index } r \text{ from } x_s^+ = \min_{i | d_i > 0} \frac{x_{B,i}}{d_i}$$

$$\mathbf{x}_B^+ = \mathbf{x}_B - \mathbf{d} x_s^+$$

New basic variables

$$\mathbf{x}_N^+ = [0, \dots, 0, x_s^+, 0, \dots, 0]^\top$$

New non-basic variables

New basis:

$$\mathcal{B} \leftarrow \mathcal{B} \setminus \{r\} \cup \{s\}, \quad \mathcal{N} \leftarrow \{1, \dots, n\} \setminus \mathcal{B}$$

Table 1: One iteration of the simplex method.

Initialization:

$$k \leftarrow 0$$

Iteration counter

$$\mathbf{x}^0, \mathbf{H}^0$$

Initial point and Hessian approx. (pos. def.)

$$\epsilon$$

Convergence tolerance

repeat

$$\mathbf{s}^k \leftarrow \text{Determine search direction}$$

$$\alpha^k \leftarrow \text{Choose step size}$$

$$\mathbf{x}^{k+1} \leftarrow \mathbf{x}^k + \alpha^k \mathbf{s}^k$$

Next iteration point

$$k \leftarrow k + 1$$

until $\|\nabla_x L(\mathbf{x}^k, \boldsymbol{\lambda}^k, \boldsymbol{\mu}^k)\| \leq \epsilon$

Or other tolerance criterion

Table 2: Structure of the SQP method.